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NBA Accredited (UG - 6 Years)

Department of Electronics & Communication Engineering

Date: 22/10/2024	**Test - 1**	**Max. Marks:50**
Semester: III	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course: Analysis and Design of Digital Circuit with HDL		**Code: EC234AI**

Part A				
Sl. No.	Questions	M	BT	CO
1.	If x=4'bX100 then &x equals____ and ^ x equals___.	1	2	1
2.	Suppose reg [127:0]y; reg [0:7]Y1; then Y1=y[25+:8] is	1	2	2
3.	The default size an integer, -------- bits.	1	2	1

4.	Consider the following verilog code segment
	`wire [5:0] A, B;`
	`wire C;`
	`assign C = ^A;`
	If the values of A and B are 5'b10011 and 5'b01110 respectively, what will be the value of {A[3:1], 2{C}, B[2:0]}?
		1	2	3

5.	What are the possible output values of == operator	1	2	1

6.	For the following Verilog code segment, if the initial value of "din" is 32'h3456ABCD, the value of "**data**" in decimal will be
	(Note that "data" is a 4-bit variable)
	wire [31:0] din;
	wire [3:0] data;
	wire [15:0] d1;
	wire [31:16] d2;
	assign d1 = din[31:16];
	assign d2 = din[15:0];
	assign data = d1[7:4] | d2[23:20] | d2[27:24];
		2	3	4

7.	If A=5'b11010, B=5'b11000, C=5'b00100. Then Y=(^(A>>>3))&&((B||C) && (B&C)) returns _____.	2	2	2
8.	For the following Verilog module the output f equal to ____. module Q1 (f, a, b, c); input a,b,c; output f, wire t1, t2; nand G1 (t1,a,b); nor G2 (t2,b,c); or G3 (f,t1,t2); endmodule	1	3	3

Part B		Maximum marks= 50		
Sl. No.	Questions	M	BT	CO
1.	a. Explain FPGA design flow with a neat diagram	5	2	1
	b. Write a short note on i. Wire and Reg ii. Bufif1 and notif0	5	1	1
2.	a. Design a 3X3 array multiplier with two 3-bit input, "a" and "b", and a 6-bit output, "p" using basic gates. Write a Verilog code using structural- level modeling for the same.	6	3	3
	b. Design and write a Verilog code for a 4 bit priority encoder with the inputs having priority 2,0,1,3	4	2	2
3.	c. Using conditional assignment operator, write a Verilog code for a full adder. Using designed full adder, write a Verilog code for 6 bit parallel adder.	7	4	4
	d. Write a Verilog code for 32 bit wide tri state buffer using data flow modelling	3	3	2
4.	a. Realize a **2-to-1 Multiplexer** using basic gates. Write a Verilog module using gate level structural description. The module (mux2x1) will take as arguments the following: 2 input bits (a,b), one selection bit (sel) and one output bit (y).Design **8-to-1 Multiplexer** using 2-to-1 multiplexer Write a Verilog module for same by instantiating required number of M**ux2x1** multiplexer declared above. The module will take as arguments the following: 8 input bits(d), 3 selection bits(s) and 1 output bit(y).	7	4	3
	b. Design a 3-bit gray to binary converter using gate level modelling	3	3	3
5.	a. Design a 2-bit comparator using 1-bit comparator and basic gates. Write a structural model of a designed 2-bit binary comparator. Develop a test bench to verify the functionality of same.	7	4	4
	b. List and explain different system tasks available in verilog	3	1	2

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	16	14	8	9	12	21		